INCH-POUND

MIL-O-60831C (AR)
12 May 1989

SUPERSEDING
MIL-O-60831B (MU)
9 August 1973

MILITARY SPECIFICATION

OBTURATING BAND, PLASTIC SPLIT (MOLDED)

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and packaging for the obturating band, plastic, split (molded) for use in the 81mm, M374 series, HE and 81mm, M375 series, WP, mortar ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Puablications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10534925 - Ring, Obturating for 81MM Mortar

(Copies of specifications, standards, drawings and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Non-government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials, shall be in accordance with the applicable specifications, drawings and this specification.

3.1.1 Certification. Certification must be furnished by the molder for each lot of rings certifying that the plastic used in molding complies with the requirements of 3.1.

3.2 Parts. The parts shall comply with the applicable requirements specified on drawing (dwg.) 10534925 and associated drawings and with all requirements specified in applicable specifications, standards, and this specification.

3.2.1 Molded parts. Obturating rings made in accordance with the applicable drawings shall be produced by the injection molding process.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. The molded parts shall be uniform in color, free of cracks, voids, burn marks, flashing, contamination, sprues and gates.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply.

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4.1.3 Tool control dimensions. Dimensions marked tool control shall be gaged during first article testing and after each 1000 items produced from each cavity mold for each machine in operation and after any tooling change. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-O-60831C (AR)

PARAGRAPH	TITLE: Ring, Obturating		SHEET OF		DRAWING NUMBER: 10534925
					NEXT HIGHER ASSEMBLY: N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ring, Obturating (Dwg. 10534925)				
	Examination for defects	100		3.1/3.2	4.4.2.1
	Certification not present	1		3.1.1	- -

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain obturated rings manufactured from one batch or "run" of resin and molded in one mold on one machine.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.1	Ring, Obturating				10534925
					NEXT HIGHER ASSEMBLY: N/A
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Width		0.40%	3.2	Gage
103	I.D. (functional)		0.40%	3.2	Gage
104	Radius of corners on outside diameter max.		0.40%	3.2	Gage
105	Certification not present	1		3.1.1	Visual
Minor					
201	Overlap		0.65%	3.4	Visual
202	Surface finish improper		0.65%	3.4	Visual (see 4.5.2)
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein starting whith paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.5 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Surface texture. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the requirements of Level C of MIL-STD-1169. In addition:

a. Bands shall be placed on cardboard tubes, the dimensions of which are as follows:

Length	- 20 inches nominal
Thickness	- 0.10 inches nominal
Outer diameter	- 2 7/8 inches $\pm$ 0.015

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The tubes shall be smooth with no delaminations, breaks or bends on the outside surface.

b. The bands shall be placed on the tubes such that no overlapping of bands shall occur.

c. Ninety to one-hundred bands shall be placed on each tube and the tubes with bands shall be placed into individual polyethylene bags and sealed.

d. The tubes shall then be placed into the compartments of a corrugated cardboard box measuring 20" by 20" by 20" high which has been partitioned with corrugated cardboard into 36 uniform compartments.

e. The boxes will then be sealed in the standard manner ready for shipment.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use on the 81mm mortar systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. See MIL-A-48078.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of (DOD) FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

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Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.4.2.1	Certification Data/ Report	UDI-A-23264B	- -

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Submission of inspection equipment designs for approval. See para 6.2.3 of MIL-A-48078. Submit designs as required to: Commander, US Army Armament, Munitions and Chemical Command (AMCCOM), ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, Data Item DI-R-1714 will apply.

6.6 Drawings. Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal and U.S. Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.7 Subject term (key word) listing.

Obturating band
Obturating ring
Plastic, split (molded)

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1395-A255)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-60831C (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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